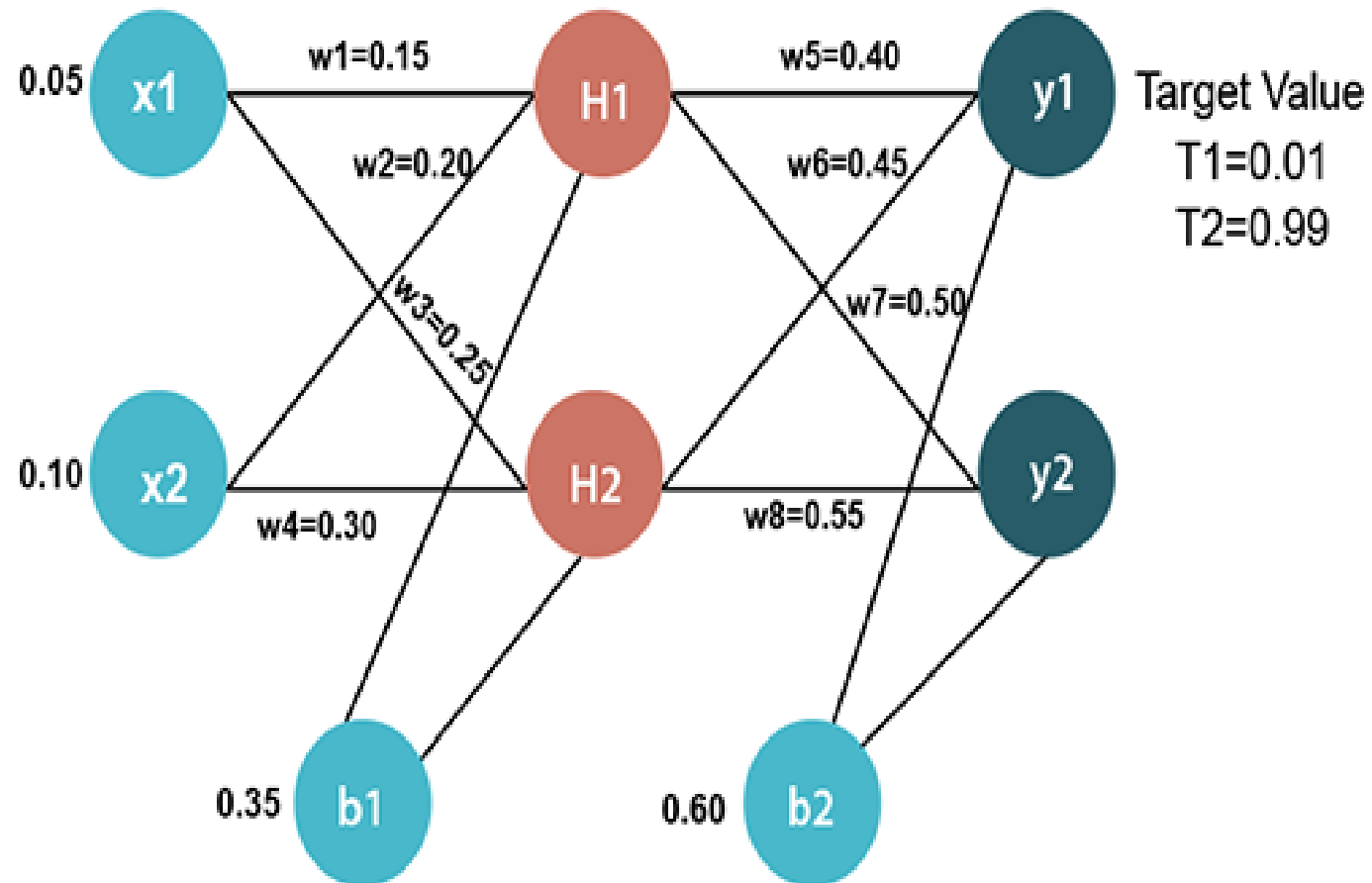


# Compute the weights



## Input values

$x_1=0.05$

$x_2=0.10$

## Initial weight

$w_1=0.15$      $w_5=0.40$

$w_2=0.20$      $w_6=0.45$

$w_3=0.25$      $w_7=0.50$

$w_4=0.30$      $w_8=0.55$

## Bias Values

$b_1=0.35$      $b_2=0.60$

## Target Values

$T_1=0.01$

$T_2=0.99$

The technique used to determine how much a weight can be changed is known as gradient descent method

Forward propagation:

$$\begin{aligned}\text{net input for } h_1 &= (x_1 \times w_1) + (x_2 \times w_2) + b_1 \\ &= [(0.05 \times 0.15) + (0.10 \times 0.20)] + 0.25 \\ &= 0.3775\end{aligned}$$

output of  $h_1$  after applying sigmoidal activation function

$$f(z) \Rightarrow \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-0.3775}} = 0.595$$

$$\boxed{h_1 \text{ output} = 0.59526992}$$

$$\begin{aligned}h_2 &= [(x_1 \times w_3) + (x_2 \times w_4)] + b_2 \\ &= [(0.05 \times 0.25) + (0.10 \times 0.30)] + 0.35 \\ &= 0.3925\end{aligned}$$

$$\begin{aligned}h_2 = f(z) &= \frac{1}{1 + e^{-0.3925}} = \frac{1}{2.480677865} \\ &= \frac{1}{1.675366346} = 0.473115621 \\ &= 0.596884\end{aligned}$$

Similarly

$$\begin{aligned}o_1 &= (w_5 * h_1) + (w_6 * h_2) + b_2 \\&= (0.40 * 0.893269992) + \\&\quad (0.45 * 0.596884378) + 0.6 \\&= 1.105905967\end{aligned}$$

$$\begin{aligned}\text{out } o_1 &= \frac{1}{1 + e^{-1.105905967}} = \frac{1}{1 + 0.33090910} \\&= \frac{1}{1.330910952}\end{aligned}$$

$$o_1 = 0.751365069$$

$$\begin{aligned}o_2 &= (w_7 * h_1) + (w_8 * h_2) + b_2 \\&= (0.5 * 0.893269992) + \\&\quad (0.55 * 0.596884378) + 0.6 \\&= 1.224921404\end{aligned}$$

$$\text{out } o_2 = \frac{1}{1 + e^{-1.224921404}}$$

$$o_2 = 0.772928465$$

target

$$T_1 = 0.01$$

$$T_2 = 0.99$$

output

$$y_1 = 0.75136507$$

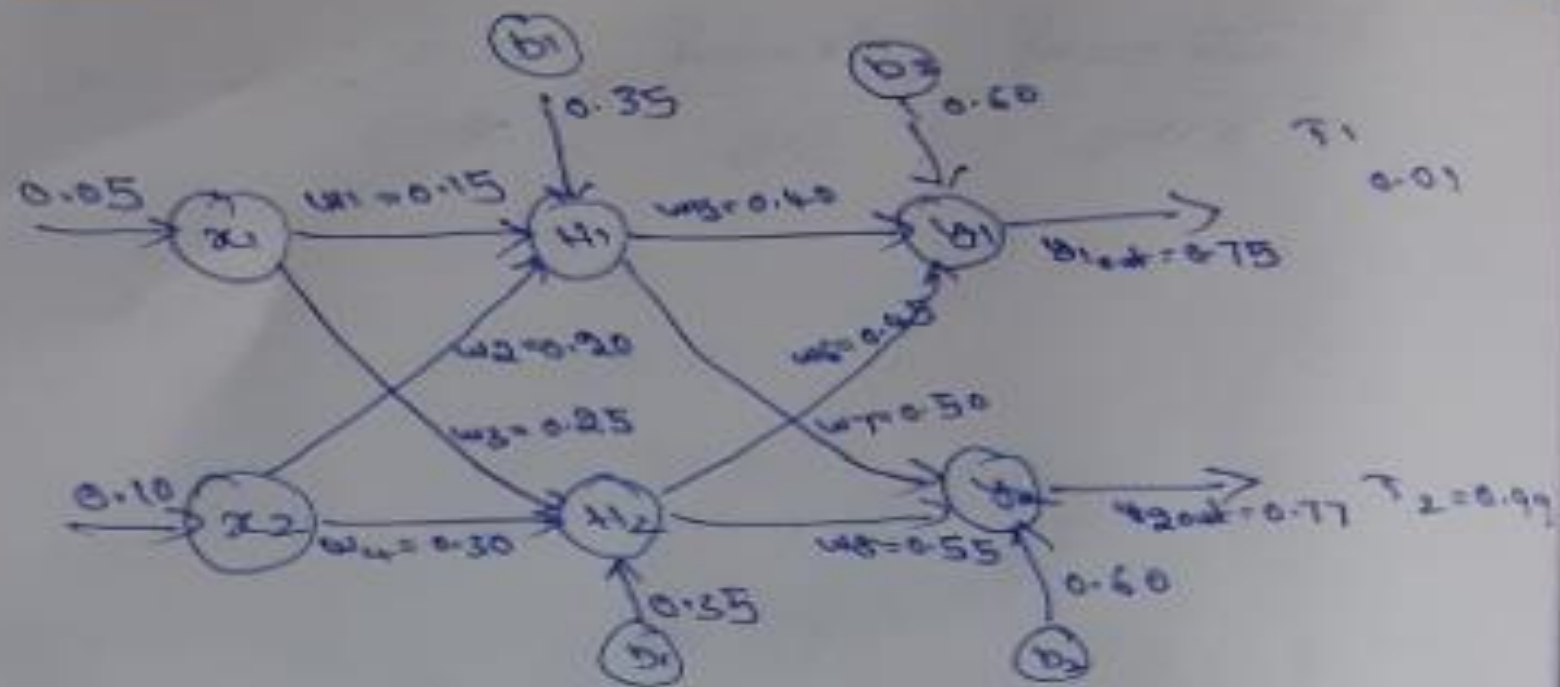
$$y_2 = 0.772928465$$

$$\text{Error} = \sum \frac{1}{2} (\text{target} - \text{output})^2$$

$$= \frac{1}{2} (0.01 - 0.75136507)^2 + \frac{1}{2} (0.99 - 0.772928465)^2$$

$$= 0.374811084 + 0.0235600257$$

$$\boxed{E_{\text{total}} = 0.29837111}$$



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$$y_1 = 1.105905967$$

$$y_2 = 1.224921404$$

$$y_{1out} = 0.751365069$$

$$y_{2out} = 0.772928465$$

compute

$$\frac{\partial E_{total}}{\partial w_5}$$

It is not possible, because  
is not present in  $E_{total}$ , so

$$\frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial y_{1,out}} \times \frac{\partial y_{1,out}}{\partial y_1} \times \frac{\partial y_1}{\partial w_5}$$

$$\frac{\partial E_{total}}{\partial y_{1,out}} = \frac{\partial \left( \frac{1}{2} (T_1 - y_{1,out})^2 + \left( \frac{1}{2} (T_2 - y_{2,out})^2 \right) \right)}{\partial y_{1,out}}$$

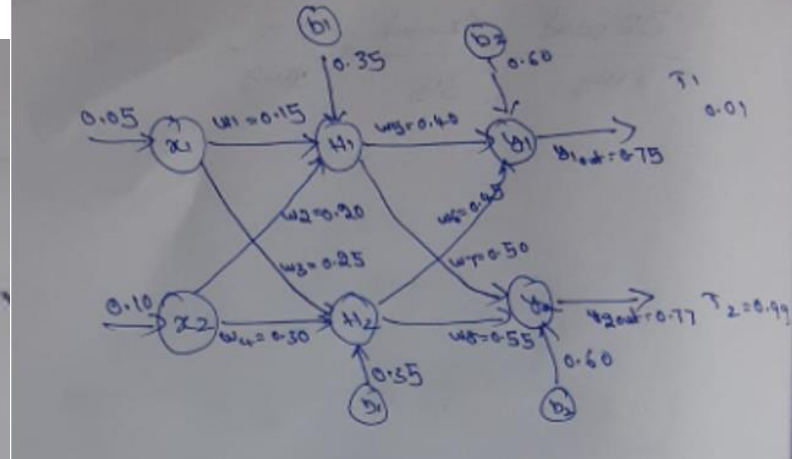
$$= 2 \times \frac{1}{2} (T_1 - y_{1,out})^{2-1} \times (0 - 1) + 0$$

$$= (T_1 - y_{1,out}) \times (-1)$$

$$= -(T_1 - y_{1,out})$$

$$= -(0.01 - 0.751365069)$$

$$\frac{\partial E_{total}}{\partial y_{1,out}} = 0.74136507$$





$$\begin{aligned}
 \frac{\partial y_{i,out}}{\partial y_i} &= \frac{\partial \left( \frac{1}{1+e^{-y_i}} \right)}{\partial y_i} \\
 &= \frac{e^{-y_i}}{(1+e^{-y_i})^2} \\
 &= e^{-y_i} \times \frac{1}{(1+e^{-y_i})^2} \\
 &= e^{-y_i} \times (y_{i,out})^2 \quad \text{--- (1)}
 \end{aligned}$$

$$y_{i,out} = \frac{1}{1+e^{-y_i}}$$

$$y_{i,out} (1+e^{-y_i}) = 1$$

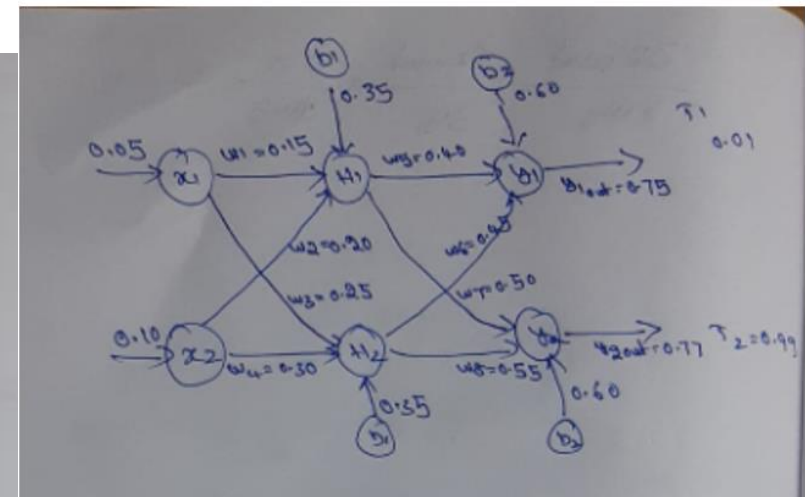
$$y_{i,out} + y_{i,out} e^{-y_i} = 1$$

$$y_{i,out} e^{-y_i} = 1 - y_{i,out}$$

$$e^{-y_i} = \frac{1 - y_{i,out}}{y_{i,out}}$$

--- (2)

replace (1) in (2)



$$\frac{\delta y_{out}}{\delta y_1} = e^{-y_1} * (y_{out})^2$$

$$= \left( \frac{1 - y_{out}}{1 + y_{out}} \right) * (y_{out})^2$$

$$\frac{\delta y_{out}}{\delta y_1} = y_{out} (1 - y_{out})$$

$$= 0.75136507 * (1 - 0.75136507)$$

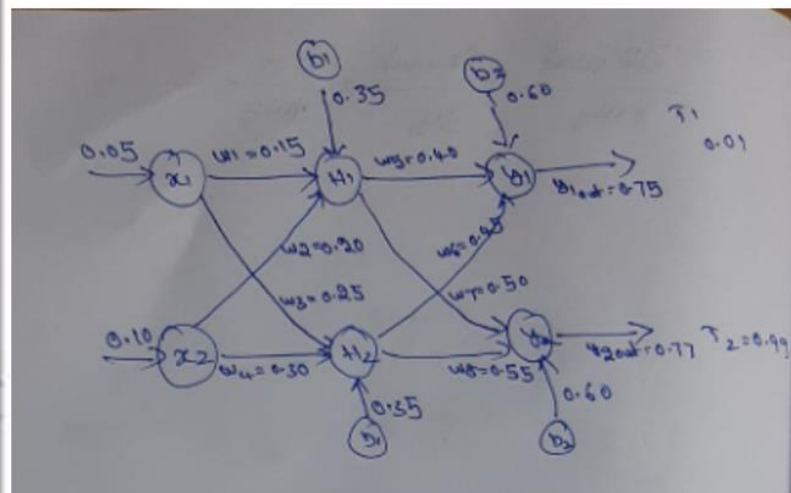
$$\frac{\delta y_{out}}{\delta y_1} = 0.186815602$$

$$\frac{\delta y_1}{\delta w_5} = \frac{\partial (H_{1out} * w_5 + H_{2out} * w_6 + D_2)}{\delta w_5}$$

Simplify

$$\frac{\delta y_1}{\delta w_5} = H_{1out}$$

$$= 0.593269992$$





So

$$\frac{S_{\text{entail}}}{S_{w5}} = \frac{S_{\text{entail}}}{S_{y,\text{entail}}} \times \frac{S_{y,\text{entail}}}{S_{y,1}} \times \frac{S_{y,1}}{S_{y,2}}$$

$$= 0.74136907 \times 0.1868 \times 0.5932$$

$$\boxed{\frac{S_{\text{entail}}}{S_{w5}} = 0.0821670407}$$

new  $w_5$  wait

$$= \boxed{w_{5,\text{new}} = w_5 - \eta \times \frac{S_{\text{entail}}}{S_{w5}}}$$

consider  $\eta = 0.5$

$$= w_{5,\text{new}} = 0.4 - 0.5 \times 0.0821670407$$

$$\boxed{w_{5,\text{new}} = 0.35891648}$$

Similarly change other weights.